

Myopia and Anchoring

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Motivation

Question: How to explain hump-shaped responses to shocks?

DSGE Literature: Add frictions or behavioral modifications to generate sluggishness

- Inconsistent with micro-level evidence

Answer: Hump-shaped responses can be rationalized by dispersed private information

- Maintains “rationality” of agents
- Consistent with micro-level evidence
- Single answer for many applications

This paper

Start with common forward-looking model:

$$a_t = \mathbb{E}_t [\varphi \xi_t + \delta a_{t+1}]$$

Add dispersed private information

- first-order uncertainty of shocks
- higher-order uncertainty of others' beliefs

Main Result: Equilibrium (under conditions) will be observationally equivalent with

$$a_t = \varphi \xi_t + \omega_f \delta \mathbb{E}_t [a_{t+1}] + \omega_b a_{t-1}$$

Applications to inflation and consumption dynamics

Literature Review

Informational frictions and higher-order uncertainty

- Morris and Shin (2001,2002); Sims (2003); Woodford (2003); Mankiw and Reis (2003, 2011); Nimark (2008, 2017), Mackowiak and Wiederholt (2009, 2015); Angeletos and La'O (2009, 2010, 2013); Huo and Takayama (2015a); Angeletos and Lian (2016); Huo and Takayama (2015b); Allen, Morris, and Shin (2006); Bacchetta and van Wincoop (2006); Kasa, Walker and Whiteman (2014); Nimark (2017); Coibion and Gorodnichenko (2012, 2015); Kumar et al (2015)

DSGE, Phillips Curves, and Micro vs Macro

- Gali and Gertler (1999); Christiano, Eichenbaum, and Evans (2005); Smets and Wouters (2007); Golosov and Lucas (2007), Midrigan (2011), Alvarez and Lippi (2014), Nakamura Steinsson (2013); Havranek, Rusnak, and Sokolova (2017); Altissimo et al. (2010)

Bounded Rationality

- Gabaix (2016); Farhi and Werning (2017); Woodford (2018); Sargent (1993); Evans and Honkapohja (2001, 2009); Marcet and Nicolini (2003); Eusepi and Preston (2016); Carvalho et al (2017)

1 Incomplete Information and Behavioral Economy Equivalence

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3 Application to Consumption

4 Conclusion

Incomplete Information Framework

Continuum of agents: $i \in [0, 1]$

Time is discrete: $t \in \{0, 1, 2, \dots\}$

Agents choose action $a_{i,t} \in \mathbb{R}$. Best response assumed to follow:

$$a_{i,t} = \mathbb{E}_{i,t}[\varphi \xi_t + \beta a_{i,t+1} + \gamma a_{t+1}]$$

$$\xi_t = \rho \xi_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, 1)$$

$$x_{i,t} = \xi_t + u_{i,t}, \quad u_{i,t} \sim \mathcal{N}(0, \sigma^2)$$

Complete Information Case ($\sigma = 0$)

All agents share same information and this is common knowledge:

$$\bar{\mathbb{E}}_t = \mathbb{E}_{i,t}, \quad a_{i,t} = a_t, \quad \delta = \beta + \gamma$$

$$a_t = \mathbb{E}[\varphi \xi_t + \delta a_{t+1}]$$

Iterating on the recursive formulation:

$$a_t = \varphi \sum_{k=0}^{\infty} \delta^k \mathbb{E}_t[\xi_{t+k}]$$

Incomplete Information ($\sigma > 0$) introduces complexity of higher-order beliefs

$$a_t = \underbrace{\sum_{k=0}^{\infty} \beta^k \bar{\mathbb{E}}_t[\varphi \xi_{t+k}]}_{\text{PE Component}} + \gamma \underbrace{\sum_{k=0}^{\infty} \beta^k \bar{\mathbb{E}}_t[a_{t+k+1}]}_{\text{GEComponent}}$$

Case 1: $\beta = \delta, \quad \gamma = 0$

$$a_t = \varphi \sum_{k=0}^{\infty} \delta^k \bar{\mathbb{E}}_t[\xi_{t+k}]$$

Case 2: $\beta = 0, \quad \gamma = \delta$

$$a_t = \varphi \sum_{k=0}^{\infty} \delta^k \bar{\mathbb{E}}_t[\bar{\mathbb{E}}_{t+1}[\cdot \cdot \cdot \bar{\mathbb{E}}_{t+k}[\xi_{t+k}] \cdot \cdot \cdot]]$$

Proposition

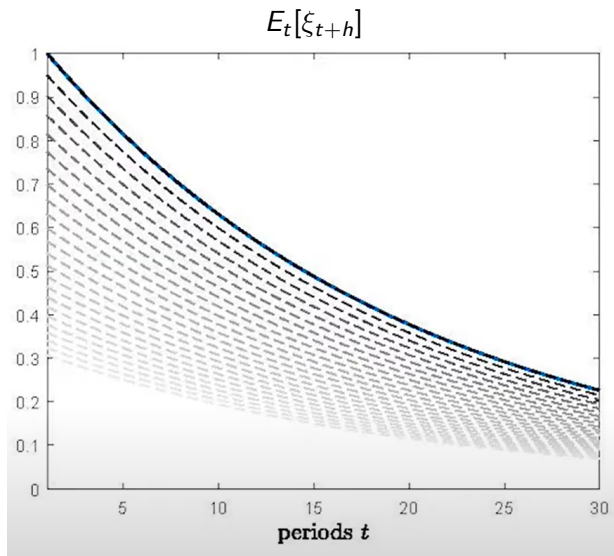
The equilibrium exists, is unique, and is such that the aggregate outcome follows the following law of motion:

$$a_t = \left(1 - \frac{\vartheta}{\rho}\right) \frac{\varphi}{(1 - \vartheta L)(1 - \rho\delta)(1 - \rho L)} \eta_t$$

where $\vartheta \in (0, \rho)$ and is given by reciprocal of largest root of a cubic $C(z)$, where $C(z)$ depends on $(\beta, \gamma, \rho, \sigma)$

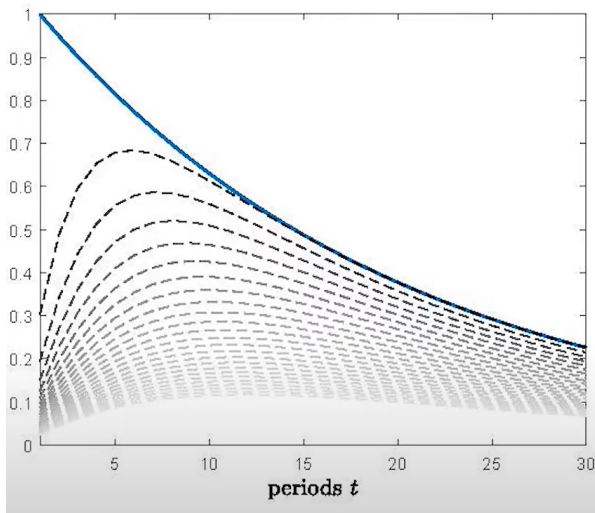
- ϑ is increasing in σ*
- ϑ is increasing in γ (holding δ fixed)*

“Higher” Order Beliefs under Complete Information



Higher Order Beliefs under Incomplete Information

$$\bar{\mathbb{E}}_t[\bar{\mathbb{E}}_{t+1}[\cdots \bar{\mathbb{E}}_{t+k}[\xi_{t+k}] \cdots]]$$



Representative Agent whose euler condition follows:

$$a_t = \varphi \xi_t + \delta \omega_f \mathbb{E}_t [a_{t+1}] + \omega_b a_{t-1}$$

$$\xi_t = \rho \xi_{t-1} + \eta_t$$

a_t follows an AR(2) whose coefficients are functions of (ω_f, ω_b) and (φ, δ, ρ)

Proposition

Fix $(\varphi, \delta, \gamma, \rho)$. For any noise level $\sigma > 0$ in the incomplete-information economy, there exists a unique pair (ω_f, ω_b) in the behavioral economy such that the two economies generate the same joint dynamics for the fundamental and aggregate outcome. Furthermore, this pair satisfies $\omega_f < 1$ (myopia) and $\omega_b > 0$ (anchoring).

- A higher σ maps to a lower ω_f and a higher ω_b*
- A higher γ (holding δ fixed) maps to a lower ω_f and a higher ω_b*

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Incomplete Information NKPC

Consider standard NKPC:

$$\pi_t = \mathbb{E}_t \left[\frac{(1 - \chi\theta)(1 - \theta)}{\theta} mc_t + \chi\pi_{t+1} \right]$$

Incorporating incomplete information, we get:

$$\pi_t = \frac{(1 - \chi\theta)(1 - \theta)}{\theta} \sum_{k=0}^{\infty} (\chi\theta)^k \bar{\mathbb{E}}_t[mc_{t+k}] + \chi(1 - \theta) \sum_{k=0}^{\infty} (\chi\theta)^k \bar{\mathbb{E}}_t[\pi_{t+k+1}]$$

Mapping to current framework:

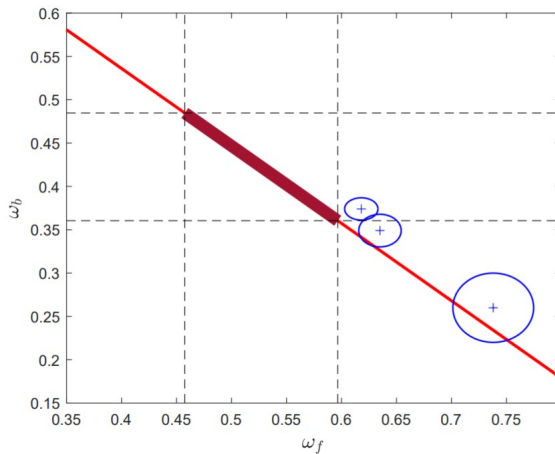
$$a_t = \pi_t, \xi_t = mc_t, \varphi = \frac{(1 - \chi\theta)(1 - \theta)}{\theta}, \beta = \chi\theta, \gamma = \chi(1 - \theta), \delta = \chi$$

Applying observational equivalence:

$$\pi_t = \frac{(1 - \chi\theta)(1 - \theta)}{\theta} mc_t + \omega_f \chi \mathbb{E}_t[\pi_{t+1}] + \omega_b \pi_{t-1}$$

Estimates of the Hybrid NKPC

$$\chi = 0.99, \theta = 0.6, \rho = 0.95$$



Evidence on expectations

Coibion and Gorodnichenko (2015) run following regression and find $K_{CG} > 0$

$$y_{t+k} - \bar{\mathbb{E}}_t[y_{t+k}] = K_{CG} \left(\bar{\mathbb{E}}_t[y_{t+k}] - \bar{\mathbb{E}}_{t-1}[y_{t+k}] \right) + v_{t+k,t}$$

Proposition

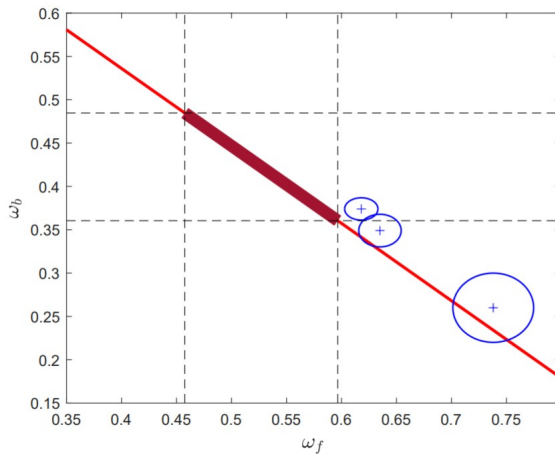
The theoretical counterpart of K_{CG} for $k = 1$ is given by:

$$K_{CG} = \lambda \frac{\vartheta + \rho - \rho\vartheta(\lambda + \vartheta) - \rho\lambda\vartheta(1 - \lambda\vartheta)}{(\rho - \lambda)(1 - \lambda\vartheta)(\rho + \vartheta - \lambda\rho\vartheta)}$$

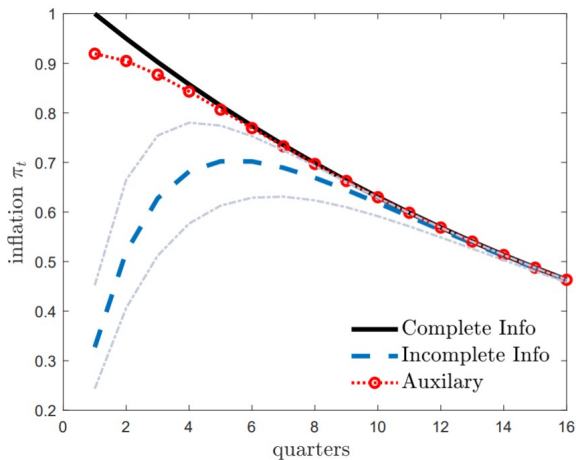
where $\lambda = \rho(1 - g)$ and g is the Kalman gain.

Estimates of the Hybrid NKPC

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Response of inflation under incomplete information



Decomposition

1 Incomplete Information and Behavioral Economy Equivalence

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Incomplete Information DIS

Consider standard DIS:

$$c_t = \mathbb{E}_t [-\varsigma r_t + c_{t+1}]$$

Incorporating incomplete information, we get:

$$c_t = -\varsigma \sum_{k=0}^{\infty} \chi^k \bar{\mathbb{E}}_t[r_{t+k}] + (1 - \chi) \sum_{k=0}^{\infty} \chi^k \bar{\mathbb{E}}_t[c_{t+k+1}]$$

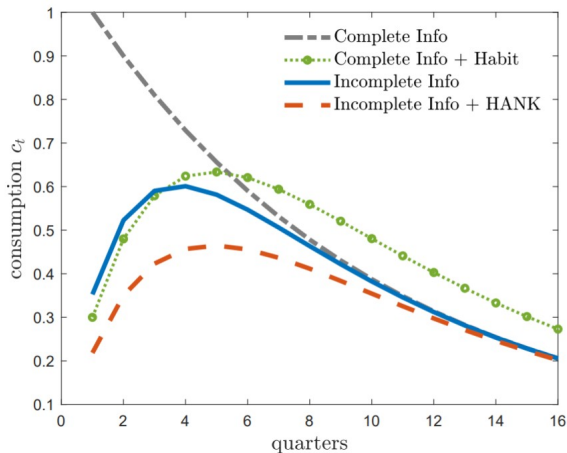
Mapping to current framework:

$$a_t = c_t, \xi_t = r_t, \varphi = -\varsigma, \beta = \chi, \gamma = 1 - \chi, \delta = 1$$

Applying observational equivalence:

$$c_t = -\varsigma r_t + \omega_f \mathbb{E}_t[c_{t+1}] + \omega_b c_{t-1}$$

Numerical Example



HANK

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Concluding Thoughts

This paper:

- Present a unified theory of hump-shaped responses that relies on incomplete information
- Prove observational equivalence with hybrid economy
- Show that theory can connect macro and micro evidence in the case of inflation

My thoughts:

- Really clean theory with many applications
- Push theory to testable predictions
- Not sure how this affects the literature going forward

Sketch of Proposition 2 proof

By the fact this is a linear gaussian model, can safely assume:

$$a_{i,t} = h(L)x_{i,t}$$

Can use Kalman filter to predict: $\mathbb{E}_{i,t}[\xi_t] = G_1(L)x_{i,t}$

Can use Kalman filter and policy rule $h(L)$ to get:

$$\mathbb{E}_{i,t}[a_{i,t+1}] = G_2(h, L)x_{i,t}$$

Can use Wiener-Hopf formula to predict: $\mathbb{E}_{i,t}[a_{t+1}] = G_3(h, L)x_{i,t}$

For policy to be consistent:

$$a_{i,t} = \mathbb{E}_{i,t}[\varphi\xi_t + \beta a_{i,t+1} + \gamma a_{t+1}]$$

$$h(L)x_{i,t} = G_1(L)x_{i,t} + G_2(h, L)x_{i,t} + G_3(h, L)x_{i,t}$$

REE equilibrium is determined by fixed point of

$$h(L) = G_1(L) + G_2(h, L) + G_3(h, L)$$

Proof of Proposition 3

Behavioral Economy aggregate outcome follows:

$$a_t = \frac{\zeta_0}{(1 - \zeta_1 L)(1 - \rho L)} \eta_t$$

$$\zeta_1 = \frac{1}{2\omega_f \delta} \left(1 - \sqrt{1 - 4\delta\omega_f\omega_b} \right), \quad \zeta_0 = \frac{\varphi\zeta_1}{\omega_b - \rho\omega_f\delta\zeta_1}$$

Incomplete Information Economy aggregate outcome follows:

$$a_t = \left(1 - \frac{\vartheta}{\rho} \right) \frac{\varphi}{(1 - \theta L)(1 - \rho\delta)(1 - \rho L)} \eta_t$$

2 equations 2 unknowns:

$$\omega_f = \frac{\delta\rho^2 - \vartheta}{\delta(\rho^2 - \vartheta^2)}, \quad \omega_b = \frac{\vartheta(1 - \delta\vartheta)\rho^2}{\rho^2 - \vartheta^2}$$

[return](#)

Decomposition Notes

$$\pi_t^* = \underbrace{\sum_{k=0}^{\infty} \beta^k \mathbb{E}_t[\varphi m c_{t+k}]}_{\text{CI PE Component}} + \gamma \underbrace{\sum_{k=0}^{\infty} \beta^k \mathbb{E}_t[\pi_{t+k+1}^*]}_{\text{CI GE Component}}$$

$$\tilde{\pi}_t = \underbrace{\sum_{k=0}^{\infty} \beta^k \bar{\mathbb{E}}_t[\varphi m c_{t+k}]}_{\text{II PE Component}} + \gamma \underbrace{\sum_{k=0}^{\infty} \beta^k \bar{\mathbb{E}}_t[\tilde{\pi}_{t+k+1}]}_{\text{II GE Component}}$$

$$\hat{\pi}_t = \underbrace{\sum_{k=0}^{\infty} \beta^k \bar{\mathbb{E}}_t[\varphi m c_{t+k}]}_{\text{II PE Component}} + \gamma \underbrace{\sum_{k=0}^{\infty} \beta^k \mathbb{E}_t[\pi_{t+k+1}^*]}_{\text{CI GE Component}}$$

return

Overlapping Generations New Keynesian Model

Standard New Keynesian Model with modifications:

- n types of agents, indexed by $g \in \{1, 2, \dots, n\}$
- π_g : mass of type g agents
- $\omega_g \in (0, 1]$: survival probability of type g
- return on saving = R_t/ω_g
- $y_{g,t} = \phi_g y_t$

Aggregate demand can be expressed as:

$$y_t = \sum_g \pi_g \left\{ -\chi \omega_g \sum_{j=0}^{\infty} (\chi \omega_g)^j \bar{\mathbb{E}}_t[r_{t+j}] + (1 - \chi \omega_g) \phi_g \sum_{j=0}^{\infty} (\chi \omega_g)^j \bar{\mathbb{E}}_t[y_{t+j}] \right\}$$

return

Aggregate Demand Equivalence Result

Proposition

(i) Under complete information, there exists a scalar $\varsigma > 0$ such that the aggregate consumption obeys a textbook Euler condition:

$$c_t = -\varsigma r_t + \mathbb{E}_t[c_{t+1}]$$

(ii) Under incomplete information, there exist scalars $\omega_f < 1$ and $\omega_b > 0$ such that aggregate consumption obeys a hybrid Euler condition of the form:

$$c_t = -\varsigma r_t + \omega_f \mathbb{E}[c_{t+1}] + \omega_b c_{t-1}$$

where ω_f and ω_b are functions of σ and $\{(\pi_g, m_g, \phi_g)\}$

Response without transfers

